

Exam Date & Time: 02-Jan-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech III Semester - Make up Examination, Jan -2025

Digital System Design and Computer Organization [ECM-2122]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Design a code converter that converts a decimal digit from 8, 4, -2, -1 code to BCD. (5)
 - A)
 - B) Explain the operation of 4 to 2 priority encoder using neat diagram and truth table. (3)
 - C) Explain the operation of accumulator-based computer organization. (2)
- 2) Design a synchronous counter with the sequence 0,1,3,7,6,4,0 using JK flip flop. (Assume unused states as don't care) (5)
 - A)
 - B) Design an FSM for two floor elevator consisting of ground and first floor. There is one button that controls the elevator, it has two values: Up or Down. Also, there are two lights in the elevator that indicate the current floor: Red for Ground, and Green for First. At each time step, the controller checks the current floor and current input, changes floors and lights in the obvious way. (3)
 - C) Implement the following function using PLA: $F=AB+BC+ABC$ (2)
- 3) Design an asynchronous 3-bit up counter using negative edge triggered JK/T flip flop and write a Verilog code for the same in data flow modelling format. (5)
 - A)
 - B) Write a Verilog code to implement 8:1 mux code using 2:1 mux. (3)
 - C) Draw a FSM diagram for a serial binary input which is divisible by 4. (2)
- 4) Use the Booth algorithm to multiply +23 (multiplicand) by -10 (multiplier), where each number is represented using 6 bits. (5)
 - A)
 - B) Explain the following addressing modes with examples: (3)
 - 1) Displacement

- 2) Indirect
- 3) Register direct

- C) Express the given statement in three instruction format: $D=A+B*C$ (2)
- 5) Explain primary modes through which data is transferred from peripheral devices to the CPU. (5)
- A)
- B) What are the different hazards in pipelining? Explain them with an example. (3)
- C) Explain static priority-based bus arbitration protocol. (2)

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